

ZHOU, CONGHAO

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CONTACT

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RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

EDUCATION

	University of California, Santa Cruz
2020 - Present	Ph.D. in Physics
2021 on Leave	Physics Department
	Thesis: <i>TBD</i>
	Committee: TESLA E. JELTEMA, ALEXIE S. LEAUTHAUD
	Duke University
2020	B.A. in Physics with Highest Distinction
	Department of Physics
	Thesis: <i>Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey</i>
	Committee: MICHAEL A. TROXEL, CHRISTOPHER W. WALTER, ROBERT L. WOLPERT

SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Constraining optical selection effect with DES and SPT, in prep.
C. ZHOU, S. GRANDIS, H.Y. WU, A.N. SALCEDO et al.
2. Resposne Bias in Shear Calibration, in prep.
C. ZHOU, M.R. BECKER, M. GATTI et al.
3. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal, in collaboration review
C. ZHOU, X. LI et al.
4. Relationship between 2D and 3D Galaxy Stellar Mass and Correlations with Halo Mass, [JCAP](#)
C. ZHOU, A.S. LEAUTHAUD, S. XU., B. DIEMER, S. HUANG, et al.
5. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data, [PRD Editors' Suggestion](#)
C. ZHOU, H.Y WU, A.N. SALCEDO, S. GRANDIS, T.E. JELTEMA, A.S. LEAUTHAUD et al.
6. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters , [MNRAS](#)
C. ZHOU, A. TONG, M.A. TROXEL, J. BLAZEK, C. LIN et al.

TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies in LSSTComCam data , [LSST Technote](#)
C. ZHOU, T. JELTEMA, A. VON DER LINDEN, L.S. KELVIN et al.

CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Vera C. Rubin Observatory Data Preview 1 , [arXiv](#)
VERA C RUBIN OBSERVATORY TEAM.
2. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters , [PRD](#)
T.N. NDE et al.
3. DESI-DR1 3×2-pt analysis: consistent cosmology across weak lensing surveys , [arXiv](#)
A. PORREDON et al.
4. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass , [arXiv](#)
K. LEIDIG et al.
5. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling , [arXiv](#)
A.N. SALCEDO et al.
6. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics , [arXiv](#)
Y. LUO et al.
7. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering , [PRD](#)
T.M.C. ABBOTT et al.
8. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1) , [arXiv](#)
C.O. CHANDLER et al.
9. Data Release 1 of the Dark Energy Spectroscopic Instrument , [arXiv](#)
DESI COLLABORATION
10. Association between optically identified galaxy clusters and the underlying dark matter halos , [PRD](#)
S. CAO et al.
11. The Outskirt Stellar Mass of Low-Redshift Massive Galaxies is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations , [ApJ](#)
S. XU et al.
12. Improving Galaxy Cluster Selection with the Outskirt Stellar Mass of Galaxies , [PRD](#)
M.KWIECIEN et al.
13. Optical galaxy cluster mock catalogs with realistic projection effects: validations with the SDSS redMaPPer clusters , [PRD](#)
A. LEE et al.
14. Building an Efficient Cluster Cosmology Software Package for Modeling Cluster Counts and Lensing , [arXiv](#)
M. AGUENA et al.
15. Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations , [PRD](#)
M. GATTI et al.

16. Dark Energy Survey Year 3 Results: Deep Field Optical + Near-Infrared Images and Catalogue , [MNRAS](#)
W.G. HARTLEY et al.

TALKS

<i>dec. 2024</i>	Theoretical Astrophysics and Cosmology Seminar, University of Arizona
<i>nov. 2024</i>	Cosmology Seminar, UC Davis
<i>apr. 2024</i>	CMB S4 Cluster AWG Meeting
<i>mar. 2024</i>	Center for the Fundamental Physics of the Universe Seminar, Brown University
<i>mar. 2024</i>	Journal Club, University of Pennsylvania
<i>oct. 2023</i>	Cosmology Seminar, Tsinghua University
<i>oct. 2023</i>	Joint Stony Brook University/ Brookhaven National Lab Cosmology Seminar

OBSERVING EXPERIENCE

<i>10.5 effective nights</i>	Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak
<i>3 nights</i>	DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO

PROFESSIONAL MEMBERSHIP

<i>2026 - present</i>	Member, Roman Science Collaboration
<i>2024 - present</i>	In-Kind Contributor, Rubin Observatory Commissioning
<i>2024 - Present</i>	Full Member, LSST Dark Energy Science Collaboration
<i>2024 - Present</i>	Student Member, American Astronomical Society
<i>2024 - Present</i>	Student Member, American Physical Society
<i>2023 - Present</i>	Member, LSST Galaxy Science Collaboration
<i>2023 - Present</i>	Member, LSST Informatics & Statistics Science Collaboration
<i>2020 - Present</i>	Sponsored Member, Dark Energy Spectroscopic Instrument
<i>2020 - 2024</i>	Member, LSST Dark Energy Science Collaboration
<i>2018 - Present</i>	Student Member, Dark Energy Survey

OUTREACH

<i>2023 - 2024</i>	Starlight Over Street Light
<i>2022</i>	UC Santa Cruz Physics Outreach Taskforce
<i>2022</i>	Science Internship Program Mentor
<i>2022</i>	UC Santa Cruz Physics Mentoring Program Mentor

PROFESSIONAL SERVICE

<i>2025 Winter</i>	UC Santa Cruz Physics Colloquium Committee
<i>2024 - present</i>	Referee, Astronomy & Astrophysics
<i>jul.2023-jun.2024</i>	Organizer, BCG Seminar, Boise State
<i>sep.2023-jun.2024</i>	Organizer, CosmoGal Seminar, UC Santa Cruz
<i>jan. 2023</i>	LOC, APS Conference for Undergraduate Women in Physics at UC Santa Cruz
<i>aug. 2022</i>	SOC, LSST DESC 2022 Chicago Meeting
<i>apr.2022- may.2025</i>	DES Early Career Scientists Committee
<i>jul. 2022 - present</i>	LSST DESC Mentoring Committee

TEACHING EXPERIENCE

<i>2024 Spring</i>	Application of Quantum Materials (Grader)
<i>2024 Winter</i>	Electricity, Magnetism, and Optics (TA)
<i>2022 Fall</i>	Quantum Mechanics (TA)
<i>2022 Summer</i>	Intermediate Physics Laboratory (TA)
<i>2022 Spring</i>	Advanced Astrophysics Laboratory (TA)
<i>3 times</i>	Introductory Physics Laboratory (TA)

April 14, 2026